

Research Paper Presentation

Attention Is All You Need

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Introduction

- This paper introduces the Transformer a new architecture for sequence tasks
- Aims to replaces recurrence and convolution with attention mechanisms
- Strengths faster training, better accuracy
- Weaknesses: Computationally complex

Methodology

- Encoder-decoder structure with self-attention layers
- Encoder: Reads input sentence into representations
- Decoder: Generates output step by step, appending to encoder's info (autoregression)
- Query = question, Key = index card describing a word, Value = the word's information (Captures relationships between all words, no matter how far apart)
- Parallel processing enables much faster training

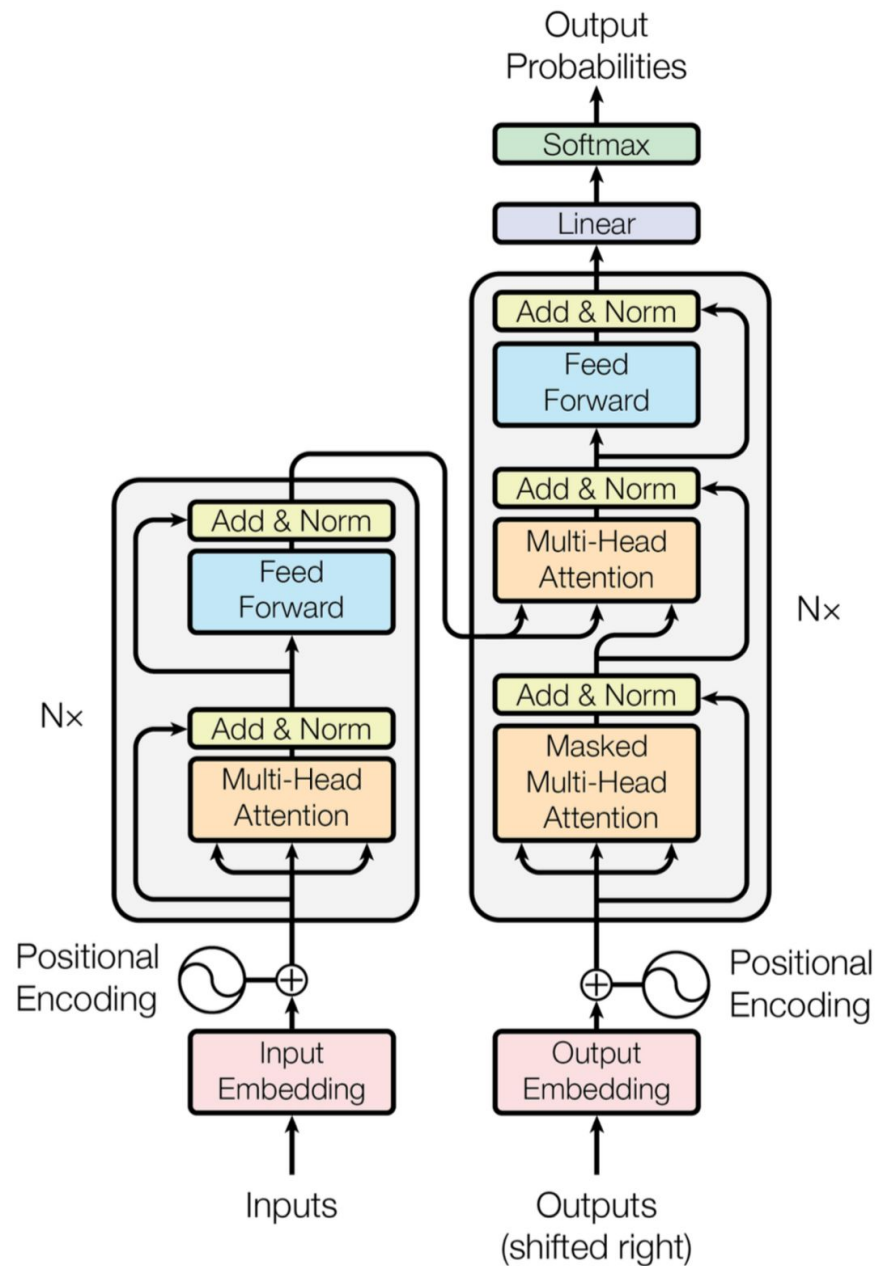


Figure 1: The Transformer - model architecture.

Attention Mechanism

- Scaled Dot-Product Attention: Matches words by similarity
- Multi-Head Attention: Looks at input from multiple perspectives
- Each Head highlights different patterns in the same input

Positional Encoding

- Adds order information since attention ignores word positions
- Sinusoid encoding uses sine and cosine waves to represent positions

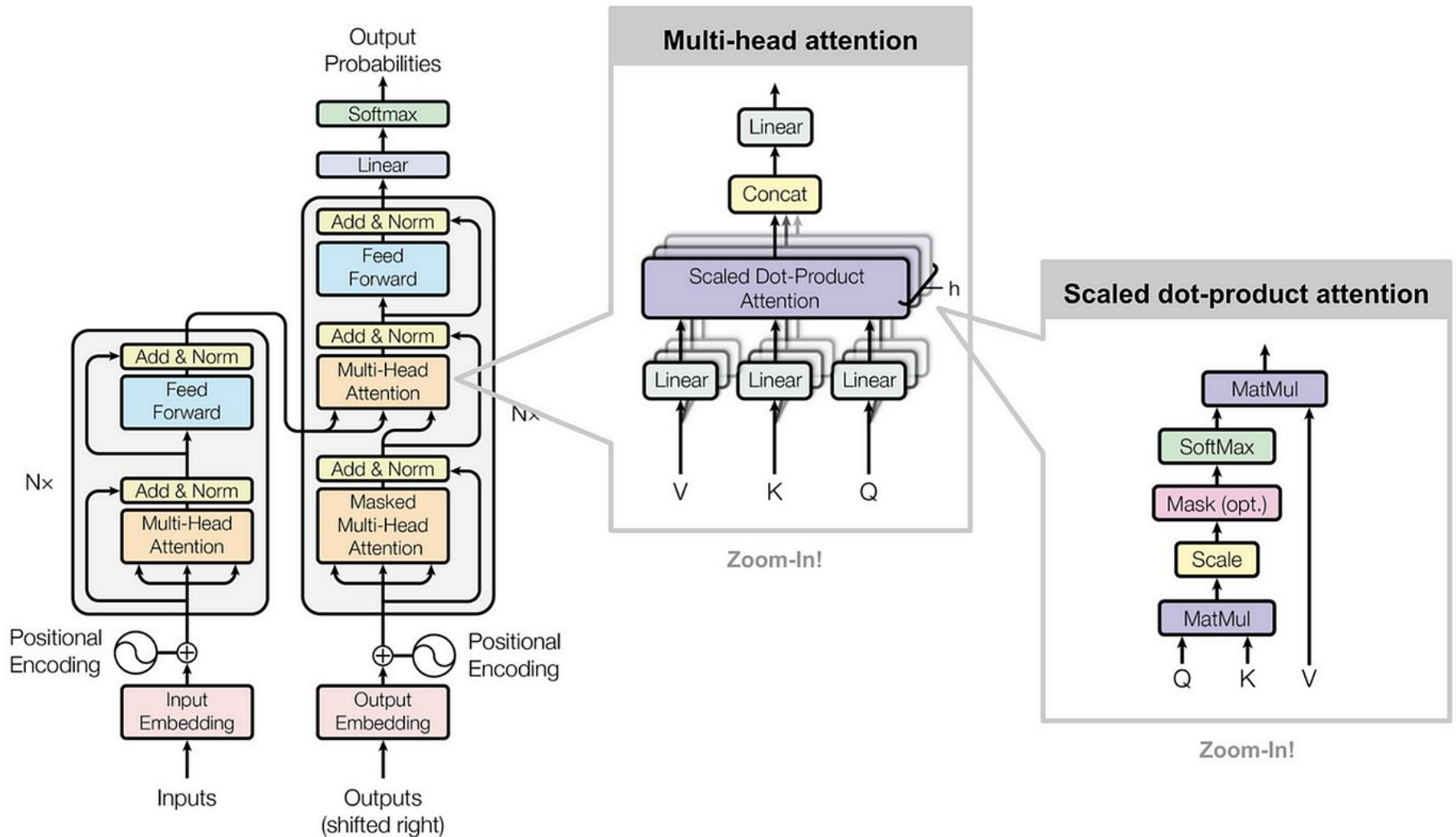
Scaled Dot-Product Attention

- Takes queries (Q), keys (K), and values (V).
- Computes attention weights using attention function (dot-product attention)
- Normalization: prevents values from becoming too large and makes training more stable.
- Produces weighted combinations of values depending on relevance between query and key.

Multi-Head Attention

- Splits queries, keys, and values into multiple “heads.”
- Each head learns different relationships between words.
- Outputs from all heads are combined and linearly transformed.

Scaled Dot-Product Attention, Multi-Head Attention & The Transformer



Why Self-Attention?

- Strengths: Shorter path for long-distance dependencies
- Strengths: Parallelizable and faster than RNNs
- Strengths: More interpretable (attention heads show focus)
- Limitations: Expensive for very long texts (quadratic cost) ($O(n^2)$ time and memory), Instead of reading one word at a time (RNN), Transformer skims whole sentence at once

Training Setup

- Data: WMT English–German (4.5M pairs), English–French (36M pairs)
- Hardware: 8 GPUs, 3.5 days of training
- Optimizer: Adam with custom learning rate schedule
- Regularization: Dropout + label smoothing

Results – Machine Translation

- Transformer (big) sets new state of the art on EN→DE (28.4 BLEU) and EN→FR (41.8 BLEU)
- Base model already better than older models at a fraction of cost
- Strength: Better accuracy + efficiency
- Limitation: Needs large compute resources for very large models

Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$	

Table 3: Variations on the Transformer architecture. Unlisted values are identical to those of the base model. All metrics are on the English-to-German translation development set, newstest2013. Listed perplexities are per-wordpiece, according to our byte-pair encoding, and should not be compared to per-word perplexities.

	N	d_{model}	d_{ff}	h	d_k	d_v	P_{drop}	ϵ_{ls}	train steps	PPL (dev)	BLEU (dev)	params $\times 10^6$
base	6	512	2048	8	64	64	0.1	0.1	100K	4.92	25.8	65
(A)				1	512	512				5.29	24.9	
				4	128	128				5.00	25.5	
				16	32	32				4.91	25.8	
				32	16	16				5.01	25.4	
(B)				16						5.16	25.1	58
				32						5.01	25.4	60
(C)	2									6.11	23.7	36
	4									5.19	25.3	50
	8									4.88	25.5	80
		256			32	32				5.75	24.5	28
		1024			128	128				4.66	26.0	168
			1024							5.12	25.4	53
(D)			4096							4.75	26.2	90
							0.0			5.77	24.6	
							0.2			4.95	25.5	
								0.0		4.67	25.3	
(E)								0.2		5.47	25.7	
										4.92	25.7	
big	6	1024	4096	16			0.3		300K	4.33	26.4	213

Model Variations

- Number of heads: too few/too many hurts performance
- Key size: smaller keys reduce accuracy
- Dropout: prevents overfitting
- Positional encodings: learned vs sinusoidal gave similar results

Generalization Beyond Translation

- Applied to English constituency parsing
- Performs well without heavy task-specific tuning
- Potential beyond translation (parsing, vision, audio)

Conclusion

- Methodology & Contribution: First model fully based on attention
- Strengths: Faster, more accurate, interpretable
- Limitations: Expensive for very long sequences, some tasks may need hybrids
- Future work: Extend to multimodal tasks (text, images, audio)

Q&A